



20 Years of TRIZ in BAE Systems

Embedding TRIZ in a Global Company

Introduction

BAE Systems are an internationally successful Global Defence and Security company employing around 100,000 people worldwide. Their wide-ranging products and services cover Air, Land and Naval forces, as well as Advanced Electronics, Security, Information Technology, and Support Services.

Oxford Creativity have been teaching TRIZ in BAE Systems for over 20 years, working with both whole teams and individuals who have registered for Oxford TRIZ public workshops. As a result, a wide mix of BAE Systems staff from all over the world learn TRIZ each year. But, for TRIZ to truly make a difference, BAE systems needed to integrate TRIZ in their company culture.

This case study explores how BAE Systems established TRIZ at a cultural level, what worked and what didn't work, highlighting the key ingredients for success and the unexpected benefits which TRIZ has delivered, on top of innovation and problem solving.

TRIZ and BAE

In 1998, Oxford Creativity's Managing Director, Karen Gadd presented TRIZ to an energetic Innovation Specialist Bob Robinson and open-minded BAE Strategy Director Pauline Marsh – they both loved everything they heard about TRIZ. In a few weeks Oxford Creativity were running their first 3-day TRIZ Workshop with BAE Systems, in the Directors Box of the Reebok Football stadium to a mixed group of Senior Management, Engineering Directors and Innovation Enthusiasts.

Since then Oxford Creativity have trained over 1,000 BAE Systems' Engineers and run Problem Solving sessions on many subjects including:

- Protection from Debris on Runways
- Location of Airports in remote Locations Management of Airports
- Design of Aircraft components
- Persuading UK Government to change/clarify the regulations for aircraft maintenance for Royal Air Force
- Innovation Culture change for processes and non-TRIZ toolkits and many, many more...

The First
3-day TRIZ
Workshop



But, as mentioned above, for TRIZ to truly make a difference, BAE Systems needed to integrate TRIZ in their company culture.

Establishing TRIZ in BAE Systems at a Cultural Level

One of the fundamental truths of TRIZ is that there are many, many possible and good solutions to most problems. Getting TRIZ accepted and embedded in large companies is just such a problem - there are many ways of succeeding and this BAE Systems story is just one.

However, if you are trying to establish TRIZ within the large organisation that you work for, you will find reading about BAE Systems' experience both familiar and helpful.

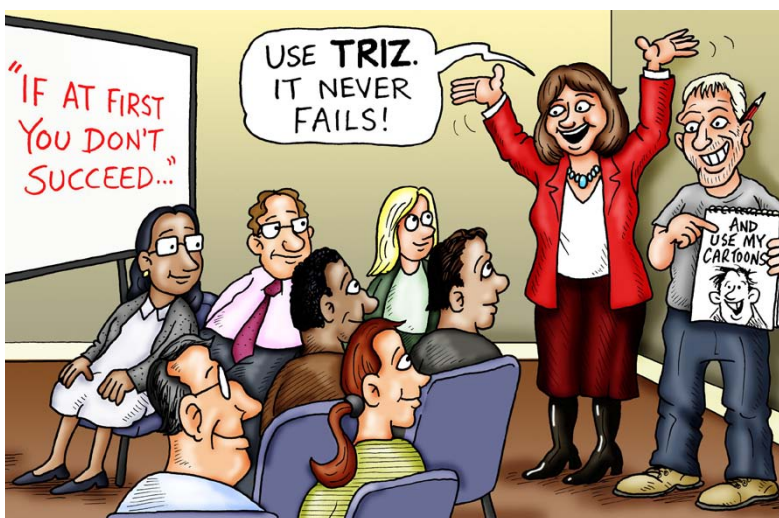
The challenges met and overcome to embed TRIZ at BAE Systems were:

1. Getting BAE Systems to accept a little-known Russian toolkit, when they were already using more famous and well-established toolkits
2. Making TRIZ easily consistently and quickly available to BAE System's 100,000 employees
3. Reaching and persuading the right people to maintain TRIZ within the organisation
4. Getting top management to appreciate the unique power of TRIZ for their challenges and commit to regular TRIZ training for everyone who wants it
5. Embedding TRIZ practices of evaluating everyone's solutions
6. Misconceptions about the complexity of TRIZ
7. TRIZ Gatekeepers
8. Misuse of TRIZ Tools

Below we look in more detail at each one and how the objections were overcome.

1. Adding to More Famous Toolkits

Unfortunately, TRIZ can be seen as just another toolkit in an overcrowded market of corporate toolkits (which range from excellent to trivial). Taking on a new toolkit makes demands on company time, money, brain power and other hard-pressed resources.



Doing this requires belief in its efficacy and power. Six Sigma and Lean are both promoted to improve efficiency and deliver cost savings and therefore widely adopted.

However, TRIZ is a unique addition to all the other toolkits, as it supplements and complements them all. The big hole in all the more famous toolkits is that they possess no direct problem solving to relevant solution concepts. In order to demonstrate this, Oxford Creativity shared with BAE several of their key articles including; Lean & TRIZ, TRIZ, Theory of Constraint & Lean, Managing Requirements with TRIZ, TRIZ Quality and Innovation.

2. Making TRIZ Accessible to Everyone

The challenge of how to make TRIZ available to everyone from one source with no new complications, extra costs or problems, is a simple one which is answered by the TRIZ Principle 'Use the resources you've got'. The resources made the most of at BAE Systems were their Life Cycle Management System (LCMS) and the BAE Systems training portals, run by Xchanging.

The Life Cycle Management System explicitly maps which tools to use when. TRIZ appears several times particularly when innovation, problem solving, and new concepts are required and Oxford Creativity's TRIZ workshops are also registered on BAE System's training systems.

This approach was a great success as it meant that no one had to explain 'What TRIZ means' in order to register for training or request problem solving. TRIZ training was available to anyone through the company-wide training portal.

Another resource used was BAE System's internal intranet. Information about TRIZ was uploaded to the intranet covering:

- TRIZ Training
- TRIZ Facilitated Problem Solving
- TRIZ Systematic approaches to new problems
- Innovation Projects
- TRIZ for Process Improvement

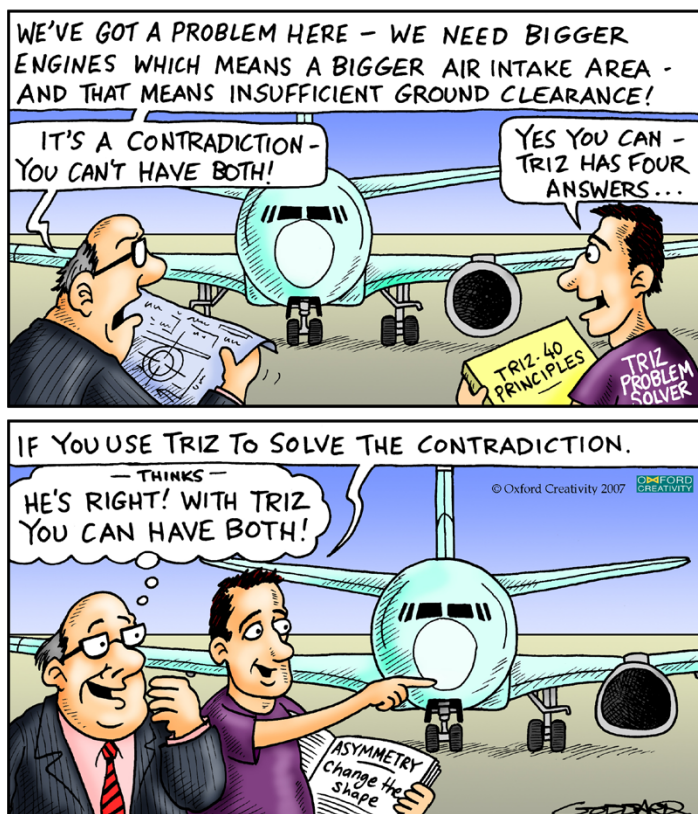
The Intranet therefore informed BAE System employees about basic TRIZ, enabled TRIZ revision, contained a template to write up case studies from TRIZ problem solving sessions, and details of TRIZ activities across the company. This meant that TRIZ enthusiasts could volunteer to join in with TRIZ problem solving outside their own areas which helped build the internal TRIZ expertise.

This led to some wonderful cross-sharing of ideas, including one lively week in Barrow when Air Systems and Submarines got together to work on each other's problems. Once they had overcome their disbelief at each other's approaches, everyone enjoyed and benefitted from sharing knowledge from their very different worlds.



3 Getting Top Management to Appreciate Unique Power of TRIZ

The challenge of persuading the budget holders, decision makers and those who have to organise the training is harder with TRIZ than say Six Sigma, which encourages everyone to claim huge savings as a result of any Six Sigma activity. TRIZ does not have this culture - although solving problems always save money.



So how was it achieved in BAE Systems?

- Oxford Creativity always ensured that they trained for practical problem solving, not theoretical TRIZ
- Workshops were made available regularly and were entirely organised by Oxford Creativity, putting no burden on anyone in BAE Systems
- The workshops were consistent and flexible, allowing people to opt in for anything from one day to five days
- Oxford TRIZ involved several carefully designed big exercises each day, for groups of 3-5 people
- All TRIZ material was clear, fun, easy to use and seen on desks for many years after the training
- Feedback was always enthusiastic
- Oxford Creativity were flexible and available at short notice - so when work was suddenly needed in Saudi Arabia or a next day problem solving workshop was required - they were nearly always able to help
- TRIZ training led to problem solving abilities especially when teams worked together

4 Getting the Engineering Population to Appreciate Problem-Solving Power of TRIZ

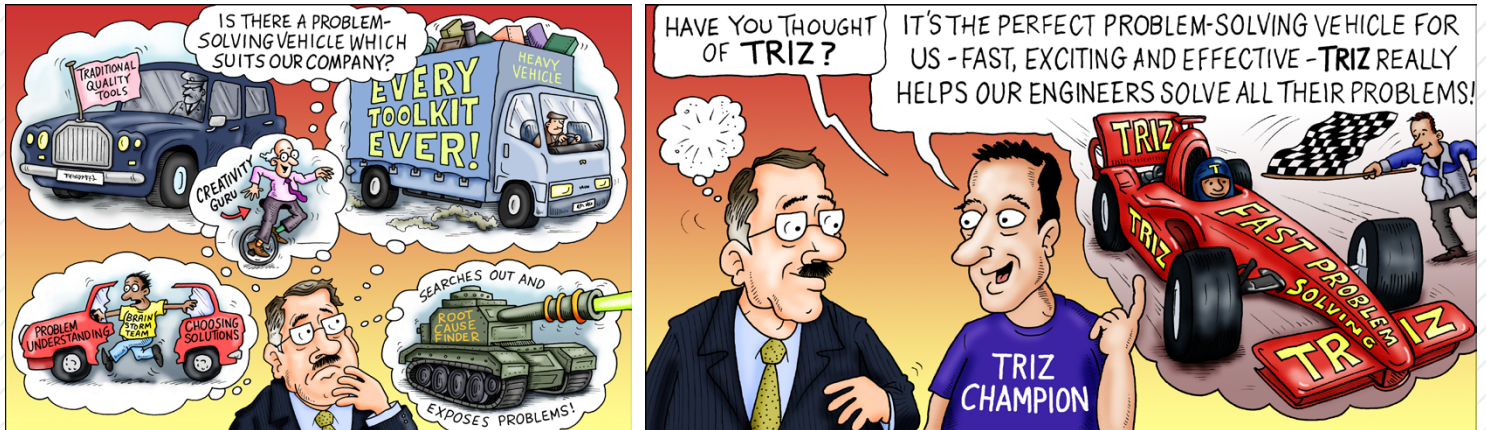
There is an English saying, 'The proof of the pudding is in the eating' - we might say that 'The proof of TRIZ is in the problem solving'. Oxford Creativity have found that helping a team solve their most difficult problems is the best way to convince people of how TRIZ will transform their thinking, their problem-solving abilities and their achievements.

In BAE Systems, Pauline Marsh invented a very ingenious, quick and efficient method for sampling TRIZ problem solving - the TRIZ Masterclass. This involved Engineers attending TRIZ training then returning to their offices and attempting to solve their most challenging problems with TRIZ. Then four or five groups of problem solvers would present their results to Oxford Creativity and BAE Systems top management.

These sessions had enough time allotted so that Oxford Creativity and BAE's management could not only judge their results, but also revisit the problem solving and

improve, expand and extend it as necessary. This proved to be very successful, motivating, good for solving difficult, previously 'unsolvable' problems and great fun.

In many other areas, problems were solved using more traditional TRIZ methods, directed and assisted by Oxford Creativity TRIZ experts. Granting sufficient time to problem solving is always seen as a luxury no matter how important or urgent the problem, and BAE Systems is no exception to this, but once TRIZ was accepted as useful and shown to be very time efficient it became easier to achieve.



5. Team Problem Solving

Persuading any company to have a consistent policy for problem solving is difficult. For major companies such as BAE Systems, problem solving is both complex and varied. It defies most simple attempts at categorisation as it is needed in all areas of business and decision making and at all stages of creating and using systems.

Problems can range from simple (easy to solve) to very difficult (require innovative solutions) and furthermore different problems need different experience and knowledge to match them to the right solutions. Brainstorming on its own often works on simple problems and sometimes works on difficult problems.



Convincing confident managers that TRIZ offers so much more than brainstorming is essential to extend and improve the problem-solving sessions. Brainstorming is wonderful and an essential first stage of the TRIZ problem solving. However, it has its pitfalls and often throws up solutions which intoxicate their owner and can set us off in the wrong direction – such situations have often been much helped by the cartoon on the left.

The culture of an Oxford TRIZ session helped to break this with much humour and goodwill on all sides, especially at its first stage of gathering all ideas from all team members and then allowing everyone to both praise and criticise them.

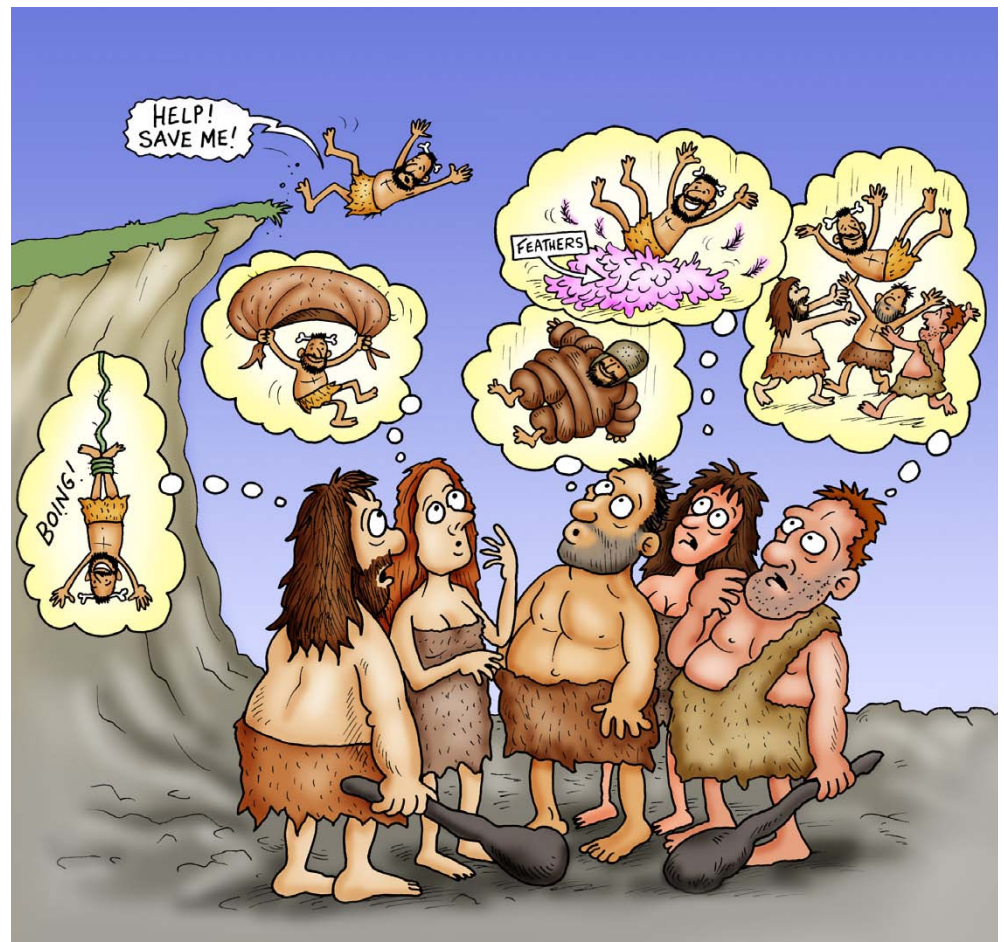
6. Misconceptions about Complexity

Initially, when promoting TRIZ in BAE Systems, there was some resistance even from the original thinkers – those few who had heard of it were hesitant as they perceived it as a complex toolkit, hard to master and apply.

This is a misapprehension, as each TRIZ tool is fairly simple and straightforward and learning the entire toolkit is not a problem for anyone. Although TRIZ is quite a large and rigorous toolkit, many of the tools overlap, as TRIZ is designed to suit all types of problem solvers. Part of the genius of TRIZ is that it was developed to allow individuals to build their own, personal TRIZ toolkit which suits their problem-solving style best.

This is a bit like having a well-equipped gym with a large range of equipment – individuals choose and use only part of that equipment depending on what suits them and which fitness problem they are tackling. The TRIZ Toolkit is much the same – there are tools which have specific purposes and tools which work in many ways on different problems and an individual finds suit them well, and which they will use extensively.

In order to combat this, Oxford Creativity made sure to always show TRIZ as simple and straightforward, easy to learn and understand. They also ensured everyone understood that when the TRIZ training finished, the TRIZ learning continues through real problem solving. By encouraging BAE employees to use TRIZ for themselves Oxford Creativity were able to prove that TRIZ was an easy and effective tool to use.



7. TRIZ Gatekeepers

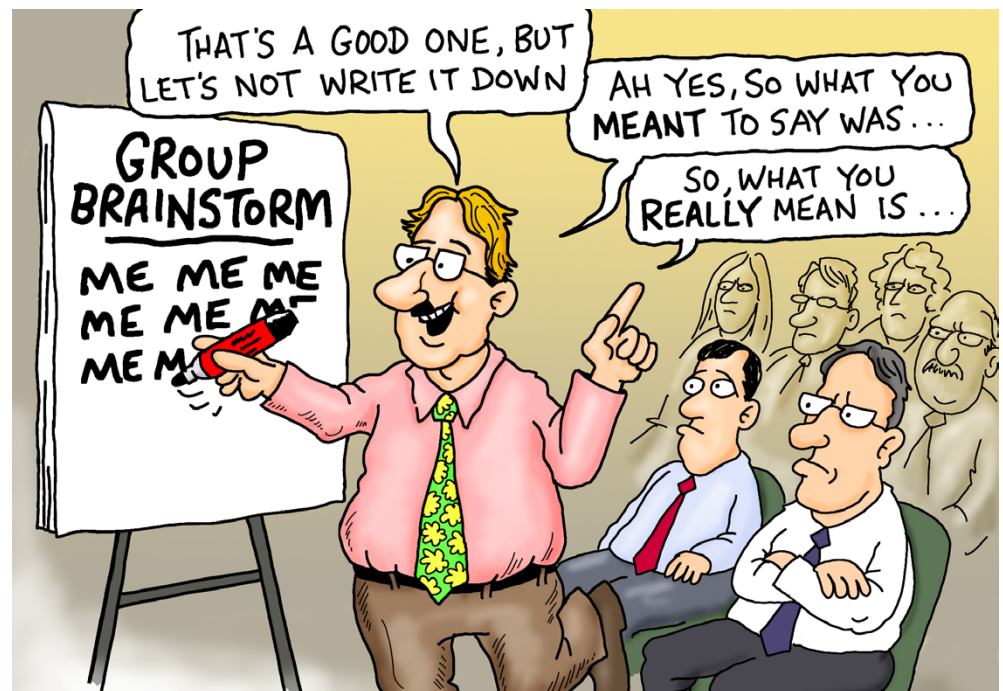
In our early days of problem solving with BAE Systems we encountered TRIZ enthusiasts who became TRIZ gatekeepers – these people love TRIZ but can hold back the widespread use of TRIZ within their own organisation.

Gatekeepers like to emphasise how hard TRIZ is, how difficult to finally master, how it will take years before they could problem solve like them.

Because Gatekeepers want to be seen as the TRIZ gurus, they often work hard before problem solving sessions and always present their solutions (arrived at through TRIZ) which awes everyone else attending.

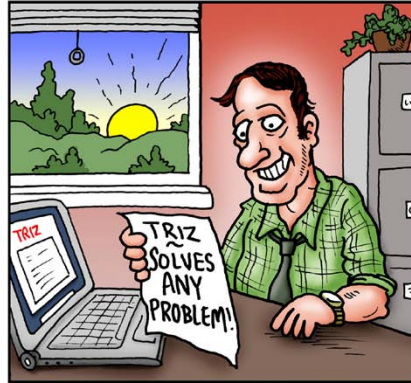
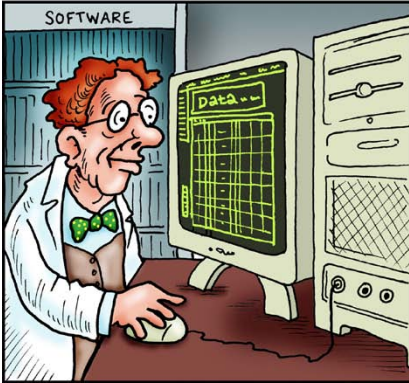
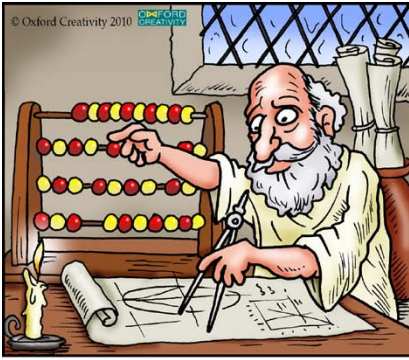
However from long experience, Oxford Creativity find that if you give the problem holder TRIZ skills that their solutions are generally better than any internal TRIZ Gatekeeper or TRIZ expert's more uninformed ideas.

One of the things that worked to prevent this at BAE, was having a large community familiar with and confident in using TRIZ, prevented a gatekeeper from “owning” TRIZ and making everyone do things his/her way.



8. Different People Like Different Tools

One great strength and problem of learning TRIZ is that individuals each like different tools and are able to develop slightly different TRIZ toolkits. The TRIZ tools are all different but some overlap and duplicate each other because they are designed to suit different styles of problem solving. Most of us will only need and will use about 80% of the tools and will reject the ones which don't suit us - which 20% we reject depends on each person.



However, it is important to learn and use all the tools so that when working in TRIZ Teams we can work together effectively, and not reject too many tools.

There is a danger of becoming familiar with a small number of the TRIZ tools and only using them – and some TRIZ gatekeepers insist that everyone else in their company limits themselves to their particular choices – which is like playing only three players in football and keeping all your best talent on the reserve bench.

In BAE Systems Oxford Creativity made sure to teach all the TRIZ tools they use for problem solving. They found that teams respond well to each other's favoured tools and took on board that understanding all the TRIZ tools - even the ones you don't favour - is essential.

Final Thoughts

This Case Study has described how Oxford Creativity worked with BAE Systems to embed TRIZ in their innovation culture, their problem-solving toolkits and their development programmes for engineers.

Whilst we know that this is just one way – others have succeeded in other companies in other ways - we hope that inspires you.

I enjoy working within large organisations with important problems to tackle, and I feel a great pride in having established TRIZ in BAE Systems.

Karen Gadd, Oxford Creativity's Managing Director

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